Probability with Engineering Applications: Week8

Piao Chen, ZJUI, Fall 2025

Programme

- Inequalities
- Joint distributions

Inequalities

Chebychev's Inequality

$$P(X \in A) = \int_A f(x)\,dx$$

Theorem:

Let X be a random variable and let $g(x)$ be a nonnegative funciton. Then, for and $r > 0$,

$$P(g(X) \geq r) \leq \frac{Eg(X)}{r}.$$

$$E[g(X)] = \int_{R} g(x) f(x)\,dx \underset{\triangle(x)}{\geq} \int_{\{x: g(x) \geq r\}} g(x) f(x)\,dx \geq r \int_{\{x: g(x) \geq r\}} f(x)\,dx$$

$g(x) \geq r$

$= r \cdot P(g(X) \geq r)$

$Y = \frac{X-\mu}{\sigma}$ $E(Y^2) = (EY)^2 + Var(Y) = 1$

Chebychev's Inequality - example

$Var(Y) = 1$

$E(Y) = E\left(\frac{X-\mu}{\sigma}\right) = \frac{1}{\sigma}E(X-\mu) = \frac{1}{\sigma}[E(X) - \mu] = 0.$

Let $g(x) = (x-\mu)^2/\sigma^2$, where $\mu = EX$ and $\sigma^2 = VarX$. For $t > 0$, we have

$$\boxed{P(|X-\mu| \geq t\sigma) \leq \boxed{\frac{1}{t^2}}.}$$

≤ 1

$r = t^2,\ t > 0.$

$$P(g(X) \geq r) \leq \frac{E\,g(X)}{r}.$$

$$\Rightarrow P\left(\frac{(X-\mu)^2}{\sigma^2} \geq r\right) \leq \frac{E\left(\frac{(X-\mu)^2}{\sigma^2}\right)}{r} = \frac{1}{r},\ \forall r > 0.$$

σ: 68%

3σ: 99.7%

Conservative bound. $\mu = 0,\ \sigma = 1,\ X \sim N(0,1),\ t = 2.$

95% : 2σ

Cheb: $P(|X| \geq 2) \leq \frac{1}{4} = 0.25$

Exact: $P(|X| \geq 2) = 1 - P(-2 < X < 2) = 0.05 << 0.25$

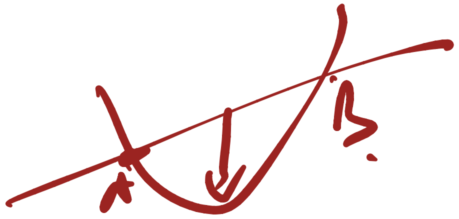
Convex vs concave

Convex: $g(\lambda x+(1-\lambda)y) \le \lambda g(x)+(1-\lambda)g(y)$
for all x and y and $0<\lambda<1$

Which of the following statements is false?

How to check:

$g''(x) \ge 0 \Rightarrow$ convex



A. The function $g(x) = 3x - 6$ is convex

$g''(x) > 0 \Rightarrow$ strictly convex.

B. The function $f(x) = \sin(x)$ is concave

C. The function $g(x) = 3x - 6$ is concave

$g''(x) \le 0 \Rightarrow$ concave

D. The function $f(x) = \sin(x)$ is not convex

$g''(x) < 0 \Rightarrow$ strictly concave.

$E(X^2) \geq (EX)^2$ ✓ $\mathrm{Var}X = E(X^2) - (EX)^2 \geq 0$

$\leq$?

Jessen's Inequality

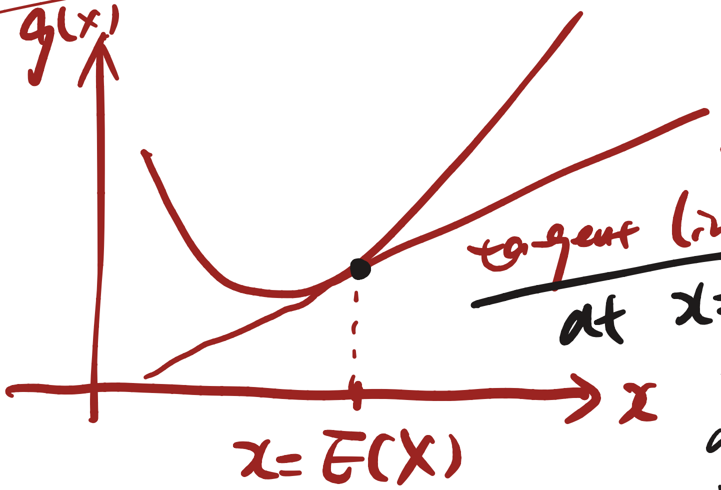
Theorem:

For any random variable X, if $g(x)$ is a convex function, then

$$Eg(X) \geq g(EX).$$

Eeuality holds if and only if, for every line $a + bx$ that is tangent to $g(x)$ at $x = EX$, $P(g(X) = a + bX) = 1$.

if $g(x)$ is Linear



tangent line at $x=EX$

$\Downarrow$

$g(EX) = l(EX)$

$l(x) = a + bx$

$\Rightarrow g(x) \geq l(x) \Rightarrow \boxed{E(g(X))} \geq E(l(X))$

$x = EX$

$= E(a + bX)$

$= a + b\,EX$

$= l(EX)$

$= \boxed{g(EX)}$

Inequalities

Let X be a random variable that is always positive. Then

$g(x) = x^2$

$g(x) = \sqrt{x}$

A. $E[X]^2 \leq E[X^2]$ and $\sqrt{E[X]} \leq E[\sqrt{X}]$

B. $E[X]^2 \geq E[X^2]$ and $\sqrt{E[X]} \leq E[\sqrt{X}]$

C. $E[X]^2 \geq E[X^2]$ and $\sqrt{E[X]} \geq E[\sqrt{X}]$

D. $E[X]^2 \leq E[X^2]$ and $\sqrt{E[X]} \geq E[\sqrt{X}]$

Jessen's Inequality

$$a_A = \frac{1}{n}(a_1 + a_2 + \cdots + a_n), \qquad \text{(Arithmetic mean)}$$

$$a_G = [a_1 a_2 \cdot \cdots \cdot a_n]^{1/n}, \qquad \text{(Geometric mean)}$$

$$a_H = \frac{1}{\frac{1}{n}\left(\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_n}\right)}. \qquad \text{(Harmonic mean)}$$

An inequality relating these means is

$$a_H \leq a_G \leq a_A.$$

① 2nd exam: Nov 28, 10:30 - 12:00, details on BB.

② cheat sheet. / handwritten.

$\boxed{\text{Week 1 - Joint distribution}}$

5/6 Qs.

$\boxed{\text{upload one}}$

② Final exam Jan 7.

2pm - 5pm

3 cheat sheet

pmf

cdf.

Joint discrete distributions

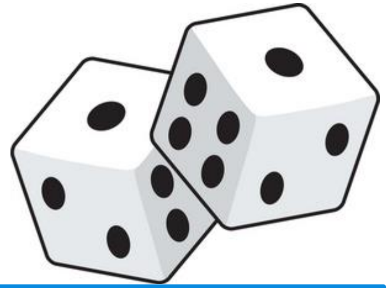
$\Omega, \mathcal{E}, P \xrightarrow{X} \mathbb{R}, \mathcal{B}, F$

toss a dice twice. $\to \Omega$ $\to$ S

$\to$ M

$\to$ Difference.

The joint probability mass function



Definition:

Let X and Y be two discrete RVs.

The joint probability mass function p of X and Y is the function defined by $p : \mathbb{R}^2 \to [0, 1]$

$$p(a, b) = P(X = a, Y = b) \text{ for all } a \text{ and } b.$$

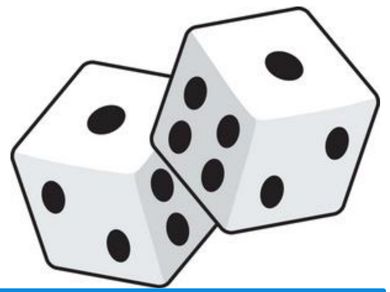
$$\sum_a \sum_b p(a,b) = 1 = P(\Omega)$$

Joint $\longrightarrow$ marginal.

$$P_X(a) = P(X=a) = \sum_b p(a,b)$$

$$P_Y(b) = P(Y=b) = \sum_a p(a,b)$$

The joint probability mass function



Definition:
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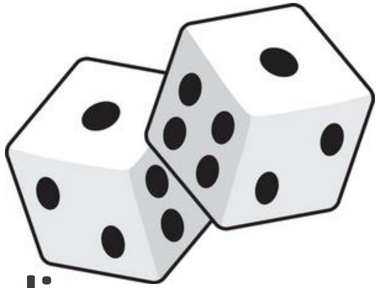
From joint to marginal:
Take sum of rows and columns. In formulas:

$$p_X(x) = \sum_y p(x, y)$$

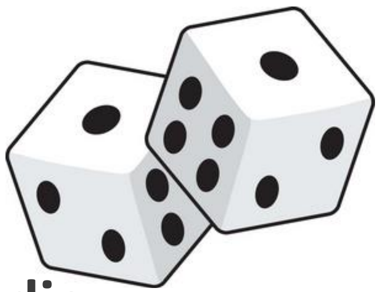
$$p_Y(y) = \sum_x p(x, y)$$

The joint probability mass function



Let S and M be the sum and maximum of two independent throws of a die.

The joint probability mass function



Let S and M be the sum and maximum of two independent throws of a die.

$S: 2, 3, \cdots 12.$

$M: 1, 2, \cdots 6$

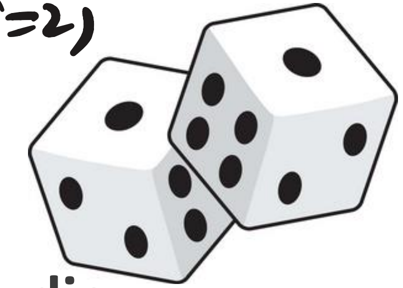
Ω

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$P(\{(i,j)\}) = \frac{1}{36}$ for any (i,j)

$P_M(1) = P(M=1)$ Joint $\rightarrow$ marginal $P_S(2) = P(S=2)$

The joint probability mass function



$P_S(3)$

Let S and M be the sum and maximum of two independent throws of a die.

$p(2,1) = P(S=2, M=1)$
$= P(\{(1,1)\}) = \frac{1}{36}$

Ω

(1,1)	(1,2)	(1,3)	(1,4)	(1,5)	(1,6)
(2,1)	(2,2)	(2,3)	(2,4)	(2,5)	(2,6)
(3,1)	(3,2)	(3,3)	(3,4)	(3,5)	(3,6)
(4,1)	(4,2)	(4,3)	(4,4)	(4,5)	(4,6)
(5,1)	(5,2)	(5,3)	(5,4)	(5,5)	(5,6)
(6,1)	(6,2)	(6,3)	(6,4)	(6,5)	(6,6)

$P(\{(i,j)\}) = \frac{1}{36}$ for any (i,j)

$P(4,3) = P(S=4, M=3) = \frac{2}{36}$
$P(6,6) = 0$

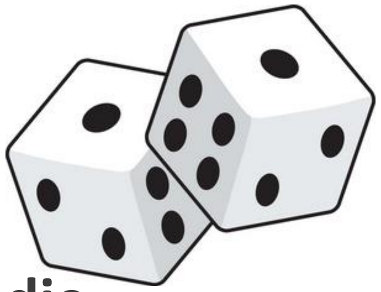
Joint probability mass function $p(a,b) = P(S=a, M=b)$.

a \ b	1	2	3	4	5	6
2	1/36	0	0	0	0	0
3	0	2/36	0	0	0	0
4	0	1/36	2/36	0	0	0
5	0	0	2/36	2/36	0	0
6	0	0	1/36	2/36	2/36	0
7	0	0	0	2/36	2/36	2/36
8	0	0	0	1/36	2/36	2/36
9	0	0	0	0	2/36	2/36
10	0	0	0	0	1/36	2/36
11	0	0	0	0	0	2/36
12	0	0	0	0	0	1/36

$\Rightarrow P_S(2) = \frac{1}{36}$
$\rightarrow P_S(3)$

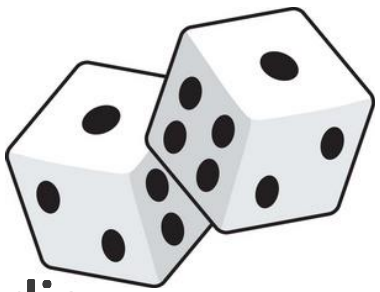
$\Rightarrow P_M(1) = \frac{1}{36}$ $P_M(2)$ $P_M(3)$ $P_M(4)$ $P_M(5)$ $P_M(6)$

The joint probability mass function



Let S and M be the sum and maximum of two independent throws of a die.

The joint probability mass function

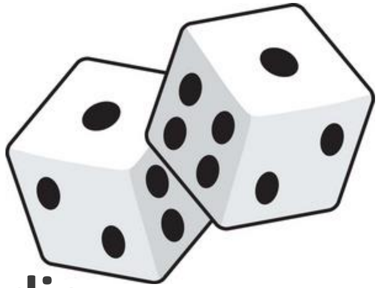


Let S and M be the sum and maximum of two independent throws of a die.

Joint probability mass function $p(a,b) = \mathrm{P}(S = a, M = b)$.

	b					
a	1	2	3	4	5	6
2	1/36	0	0	0	0	0
3	0	2/36	0	0	0	0
4	0	1/36	2/36	0	0	0
5	0	0	2/36	2/36	0	0
6	0	0	1/36	2/36	2/36	0
7	0	0	0	2/36	2/36	2/36
8	0	0	0	1/36	2/36	2/36
9	0	0	0	0	2/36	2/36
10	0	0	0	0	1/36	2/36
11	0	0	0	0	0	2/36
12	0	0	0	0	0	1/36

The joint probability mass function



Let S and M be the sum and maximum of two independent throws of a die.

Joint probability mass function $p(a, b) = \mathrm{P}(S = a, M = b)$.

	b					
a	1	2	3	4	5	6
2	1/36	0	0	0	0	0
3	0	2/36	0	0	0	0
4	0	1/36	2/36	0	0	0
5	0	0	2/36	2/36	0	0
6	0	0	1/36	2/36	2/36	0
7	0	0	0	2/36	2/36	2/36
8	0	0	0	1/36	2/36	2/36
9	0	0	0	0	2/36	2/36
10	0	0	0	0	1/36	2/36
11	0	0	0	0	0	2/36
12	0	0	0	0	0	1/36

$\Rightarrow$

	b						
a	1	2	3	4	5	6	$p_S(a)$
2	1/36	0	0	0	0	0	1/36
3	0	2/36	0	0	0	0	2/36
4	0	1/36	2/36	0	0	0	3/36
5	0	0	2/36	2/36	0	0	4/36
6	0	0	1/36	2/36	2/36	0	5/36
7	0	0	0	2/36	2/36	2/36	6/36
8	0	0	0	1/36	2/36	2/36	5/36
9	0	0	0	0	2/36	2/36	4/36
10	0	0	0	0	1/36	2/36	3/36
11	0	0	0	0	0	2/36	2/36
12	0	0	0	0	0	1/36	1/36
$p_M(b)$	1/36	3/36	5/36	7/36	9/36	11/36	1

When is a Function a Joint PMF?

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A function p is a joint pmf if and only if

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A function p is a joint pmf if and only if

1. The function p is nonnegative

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A function p is a joint pmf if and only if

1. The function p is nonnegative

2. There are finite or countably infinite sets $U = \{u_1, u_2, \dots\}$ and $V = \{v_1, v_2, \dots\}$ such that $p(u, v) = 0$ if $u \notin U$ or $v \notin V$

When is a Function a Joint PMF?

A function p is a joint pmf if and only if

1. The function p is nonnegative

2. There are finite or countably infinite sets $U = \{u_1, u_2, \ldots\}$ and $V = \{v_1, v_2, \ldots\}$ such that $p(u, v) = 0$ if $u \notin U$ or $v \notin V$

3. $\sum_i \sum_j p(u_i, v_j) = 1$

Conditional PMF

$P(A|B)$

Conditional random variable

$Y | X = x_0 \sim Y.$

The conditional pmf of Y given X is

Conditional PMF

$$P(B|A) = \frac{P(AB)}{P(A)}$$

The conditional pmf of Y given X is

$$P_{Y|X}(v|u_0) = P\{Y = v \mid X = u_0\}$$

$$= \begin{cases} \dfrac{P_{X,Y}(u_0, v)}{P_X(u_0)}, & P_X(u_0) \neq 0 \\ \text{undefined}, & P_X(u_0) = 0 \end{cases}$$

Joint PMF Example

Let (X, Y) have joint pmf as shown below

	$X = 1$	$X = 2$	$X = 3$
$Y = 1$	0	0.3	0.1
$Y = 2$	0	0.2	0.2
$Y = 3$	0.1	0.1	0

Does the function above satisfy the conditions to be a joint pmf?

1) Nonnegative?
2) Finite or countably infinite values with nonzero probability mass?
3) Probability masses sum to 1?

Joint PMF Example

$$\frac{P(2,v)}{P_X(2)} = \frac{P(2,v)}{0.3+0.2+0.1}$$

Let (X, Y) have joint pmf as shown below

	$X = 1$	$X = 2$	$X = 3$
$Y = 1$	0	0.3	0.1
$Y = 2$	0	0.2	0.2
$Y = 3$	0.1	0.1	0

$\| \ P_X(2)$

a) What is the marginal pmf of X?

b) What is the marginal pmf of Y?

c) $P\{X = Y\}$

d) $P\{X > Y\}$

e) $P_{Y|X}(v \mid u_0 = 2) = P(Y = v \mid X = 2)$

$$P(Y=1 \mid X=2) = \frac{0.3}{0.3+0.2+0.1}$$

$+$

$$P(Y=2 \mid X=2) = \frac{0.2}{0.3+0.2+0.1}$$

$+$

$$P(Y=3 \mid X=2) = \frac{0.1}{0.3+0.2+0.1}$$

$// \quad 1$

Joint PMF Example

Let (X, Y) have joint pmf as shown below

	$X = 1$	$X = 2$	$X = 3$
$Y = 1$	0	0.3	0.1
$Y = 2$	0	0.2	0.2
$Y = 3$	0.1	0.1	0

a) What is the marginal pmf of X?

Joint PMF Example

Let (X, Y) have joint pmf as shown below

	$X = 1$	$X = 2$	$X = 3$
$Y = 1$	0	0.3	0.1
$Y = 2$	0	0.2	0.2
$Y = 3$	0.1	0.1	0

b) What is the marginal pmf of Y?

Joint PMF Example

Let (X, Y) have joint pmf as shown below

	$X = 1$	$X = 2$	$X = 3$
$Y = 1$	0	0.3	0.1
$Y = 2$	0	0.2	0.2
$Y = 3$	0.1	0.1	0

$c)\ \ P\{X = Y\} =$

Joint PMF Example

Let (X, Y) have joint pmf as shown below

	$X = 1$	$X = 2$	$X = 3$
$Y = 1$	0	0.3	0.1
$Y = 2$	0	0.2	0.2
$Y = 3$	0.1	0.1	0

$d)\ \ P\{X > Y\} =$

Joint PMF Example

Let (X, Y) have joint pmf as shown below

	$X = 1$	$X = 2$	$X = 3$
$Y = 1$	0	0.3	0.1
$Y = 2$	0	0.2	0.2
$Y = 3$	0.1	0.1	0

$e)$ $P_{Y|X}(v \mid u_0 = 2) =$

Joint PMF Example 2

$(X,Y) \to (X,Z) \to Z$

$\to (Y,Z) \nearrow$

Suppose the joint pmf of X and Y is

$$P(X=i, Y=j) = \frac{\binom{j}{i} e^{-2\lambda} \lambda^j}{j!}, \quad \boxed{0 \le i \le j}, \quad \boxed{j \ge 0}$$

Determine:

a) The pmf of Y $\Rightarrow \sum_{i=0}^{j} p(X=i, Y=j)$

b) The pmf of X $\Rightarrow \sum_{j=i}^{\infty} p(X=i, Y=j)$

c) The pmf of $Y - X$

$Z = Y - X$

Joint PMF Example 2

Binomial Thm

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

a) Marginal pmf of Y:

$$P(Y=j) = \sum_{i=0}^{j} \binom{j}{i} \boxed{\frac{e^{-2\lambda}\lambda^j}{j!}}$$

$$= \frac{e^{-2\lambda}\lambda^j}{j!} \sum_{i=0}^{j} \binom{j}{i} \times 1^{j-i} \times 1^i$$

$$= \frac{e^{-2\lambda}\lambda^j}{j!} (1+1)^j = e^{-2\lambda} \frac{(2\lambda)^j}{j!}$$

$$\boxed{j \geq 0.}$$

Joint PMF Example 2

$$e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$$

b) Marginal pmf of X:

$$P(X=i) = \sum_{j=i}^{\infty} \binom{j}{i} e^{-2\lambda} \frac{\lambda^j}{j!}$$

$$= e^{-2\lambda} \sum_{j=i}^{\infty} \binom{j}{i} \frac{\lambda^j}{j!} = e^{-2\lambda} \sum_{j=i}^{\infty} \frac{\cancel{j!}}{i!(j-i)!} \cdot \frac{\lambda^j}{\cancel{j!}}$$

$$= e^{-2\lambda} \sum_{j=i}^{\infty} \frac{\lambda^j}{i!(j-i)!} = \frac{e^{-2\lambda}}{i!} \sum_{j=i}^{\infty} \frac{\lambda^j}{(j-i)!}$$

Define $k = j - i$

$\Leftrightarrow j = k + i$

$$= \frac{e^{-2\lambda}}{i!} \sum_{k=0}^{\infty} \frac{\lambda^{k+i}}{k!} = \frac{e^{-2\lambda}}{i!} \sum_{k=0}^{\infty} \frac{\lambda^k \cdot \lambda^i}{k!}$$

$$= \frac{\lambda^i \cdot e^{-2\lambda}}{i!} \boxed{\sum_{k=0}^{\infty} \frac{\lambda^k}{k!}} = \frac{\lambda^i}{i!} \cdot e^{-\lambda}, \quad i \geq 0$$

Joint PMF Example 2

$Z = Y - X$

c) What is the pmf of $\mathrm{Y} - X$?

$$P(X = i, Z = k) = P(X = i, Y - X = k).$$

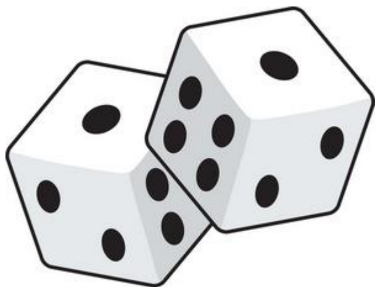
$$= \underline{P(X = i, Y = i + k)}$$

$$= \cdots$$

$$\boxed{i \geq 0,\ k \geq 0}$$

$$P(Z = k) = \sum_i P(X = i, Z = k) = \cdots$$

Joint distribution function

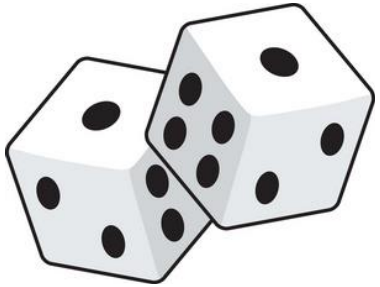


Definition:
Let X and Y be two RVs.
The joint distribution function F of X and Y is the function $F : \mathbb{R}^2 \to [0, 1]$ defined by

$$F(a, b) = P(X \leq a, Y \leq b) \text{ for all } a \text{ and } b.$$

$F_{X,Y}(a,b)$

Joint distribution function



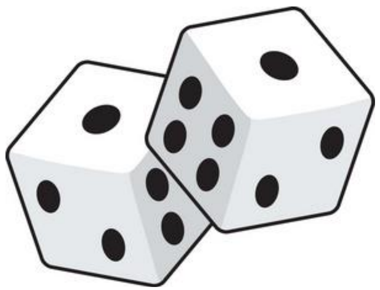
Joint probability mass function $p(a, b) = \mathrm{P}(S = a, M = b)$.

	b					
a	1	2	3	4	5	6
2	1/36	0	0	0	0	0
3	0	2/36	0	0	0	0
4	0	1/36	2/36	0	0	0
5	0	0	2/36	2/36	0	0
6	0	0	1/36	2/36	2/36	0
7	0	0	0	2/36	2/36	2/36
8	0	0	0	1/36	2/36	2/36
9	0	0	0	0	2/36	2/36
10	0	0	0	0	1/36	2/36
11	0	0	0	0	0	2/36
12	0	0	0	0	0	1/36

$F(4,3) = P(S \le 4, M \le 3)$

$F(7,5)$

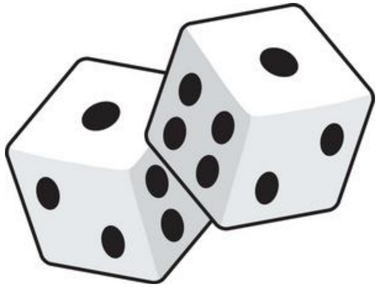
Joint distribution function



Joint probability mass function $p(a, b) = \mathrm{P}(S = a, M = b)$.

	b					
a	1	2	3	4	5	6
2	1/36	0	0	0	0	0
3	0	2/36	0	0	0	0
4	0	1/36	2/36	0	0	0
5	0	0	2/36	2/36	0	0
6	0	0	1/36	2/36	2/36	0
7	0	0	0	2/36	2/36	2/36
8	0	0	0	1/36	2/36	2/36
9	0	0	0	0	2/36	2/36
10	0	0	0	0	1/36	2/36
11	0	0	0	0	0	2/36
12	0	0	0	0	0	1/36

Joint distribution function

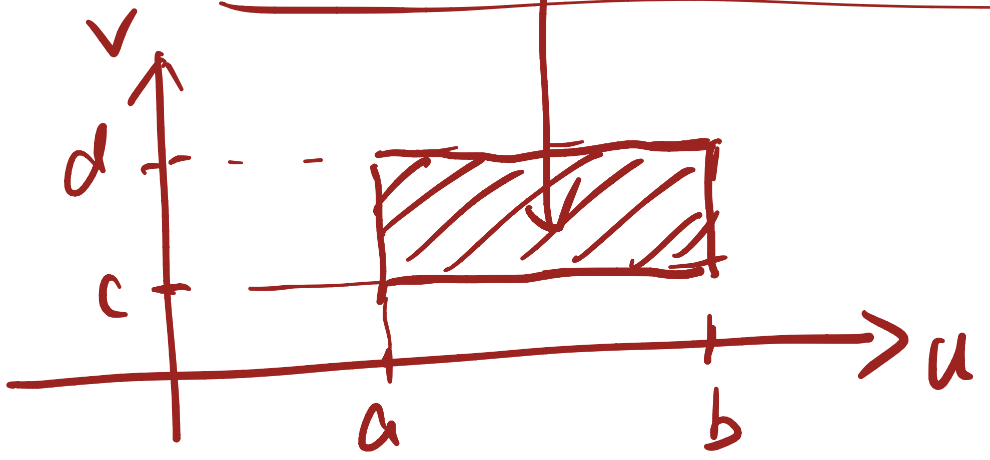


Joint probability mass function $p(a, b) = \mathrm{P}(S = a, M = b)$.

a \ b	1	2	3	4	5	6
2	1/36	0	0	0	0	0
3	0	2/36	0	0	0	0
4	0	1/36	2/36	0	0	0
5	0	0	2/36	2/36	0	0
6	0	0	1/36	2/36	2/36	0
7	0	0	0	2/36	2/36	2/36
8	0	0	0	1/36	2/36	2/36
9	0	0	0	0	2/36	2/36
10	0	0	0	0	1/36	2/36
11	0	0	0	0	0	2/36
12	0	0	0	0	0	1/36

Calculating Probability for a Rectangular Region

Consider a rectangular region: $R = \{(u, v): a < u \leq b, c < v \leq d\}$



The probability that the random point (X, Y) is in the region R is

$$P((X,Y)\in R) = F(b,d) - F(a,d) - F(b,c) + F(a,c)$$

Marginal Distributions

$$F_Y(v) = \lim_{u \to \infty} F_{X,Y}(u,v)$$

The joint CDF of X and Y can be used to determine the distributions of X and Y alone, i.e., the marginal distributions of X and Y

$$\lim_{v \to \infty} F_{X,Y}(u,v) \Rightarrow F_X(u)$$

$$\boxed{\begin{aligned} F_X(u) &= P(X \le u) = P(X \le u, Y < \infty) \\ &= F_{X,Y}(u, \infty) \end{aligned}}$$

Marginal Distributions

The joint CDF of X and Y can be used to determine the distributions of X and Y alone, i.e., the marginal distributions of X and Y

Prop. Suppose X and Y have joint CDF $F_{X,Y}$. Then,

$$F_X(u) = \lim_{v \to \infty} F_{X,Y}(u, v)$$
$$F_Y(v) = \lim_{u \to \infty} F_{X,Y}(u, v)$$

When is a function a joint CDF?

① non-decreasing.

② right-continuous.

③ lim 0 , 1.

When is a function a joint CDF?

$= F(b) - F(a) \geq 0$

$\boxed{P(a < X \leq b) \geq 0}$

univariate cdf case: ① non-decreasing ② right-continuous ③ limit

A function F is the joint CDF of a pair of random variables (X, Y) if and only if

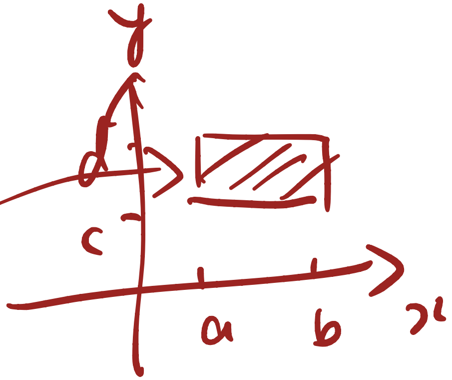
③ *1.* $0 \leq F(u, v) \leq 1 \ \forall \ (u, v) \in \mathbb{R}^2$

①② *2.* $F(u, v)$ is nondecreasing in u and nondecreasing in v

3. $F(u, v)$ is right-continuous in u and right-continuous in v

4. If $a < b$ and $c < d$, then

$$F_{X,Y}(b, d) - F_{X,Y}(b, c) - F_{X,Y}(a, d) + F_{X,Y}(a, c) \geq 0$$



③ *5.* $\lim_{u \to -\infty} F(u, v) = 0$ for each v and $\lim_{v \to -\infty} F(u, v) = 0$ for each u

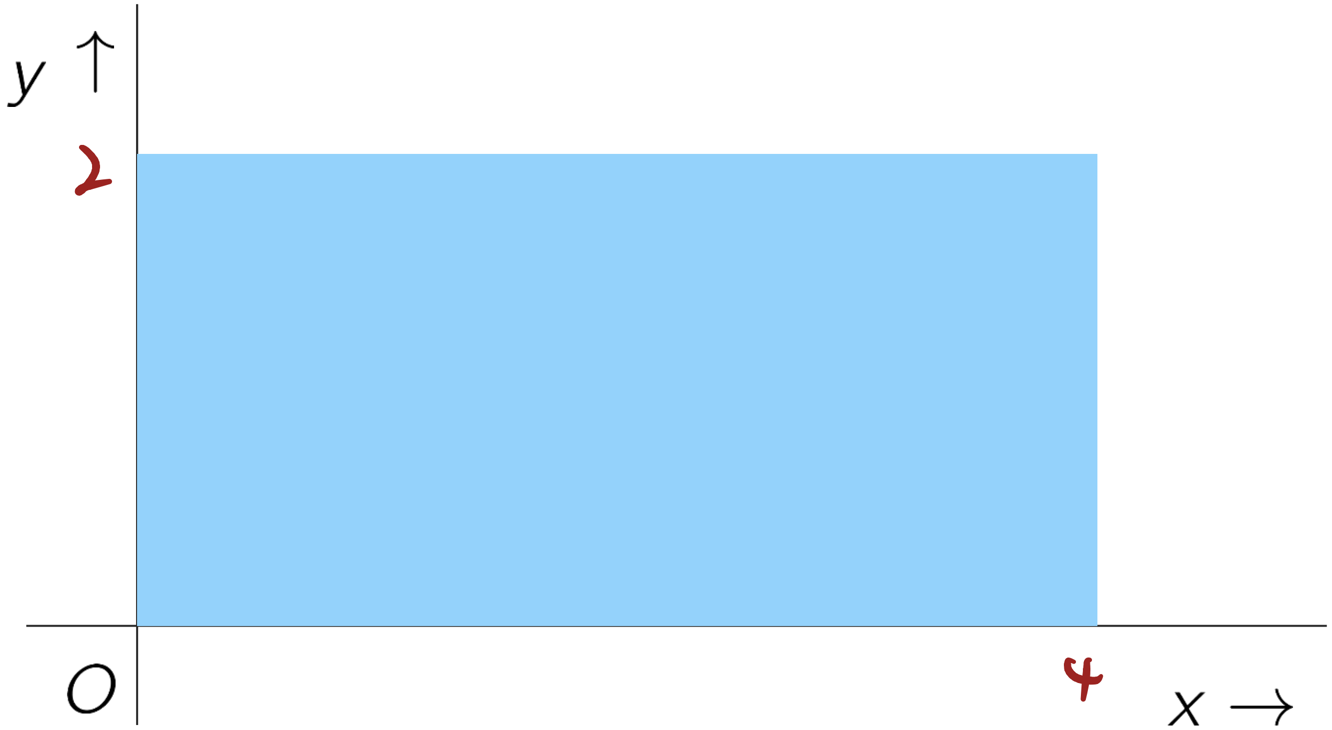
6. $\lim_{u \to \infty} \lim_{v \to \infty} F(u, v) = 1$

Joint continuous distribution

Finding a coin in the pond

$X \sim U(0,4)$.
$Y \sim U(0,2)$.

Suppose you lost a coin in a 2 by 4 rectangular pond.
You have no reason to suspect it in any particular location in the pond.



Finding a coin in the pond

Suppose you lost a coin in a 2 by 4 rectangular pond.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin

Let Y be the y-coordinate of the coin

Finding a coin in the pond

Suppose you lost a coin in a 2 by 4 rectangular pond.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin $\longrightarrow$ $X \sim U(0, 4)$

Let Y be the y-coordinate of the coin

Finding a coin in the pond

Suppose you lost a coin in a 2 by 4 rectangular pond.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin ⟶ $X \sim U(0,4)$

Let Y be the y-coordinate of the coin ⟶ $Y \sim U(0,2)$

Finding a coin in the pond

Suppose you lost a coin in a 2 by 4 rectangular pond.
You have no reason to suspect it in any particular location in the pond.

$P(X=2)=0.$

Let X be the x-coordinate of the coin → $X \sim U(0,4)$

Let Y be the y-coordinate of the coin → $Y \sim U(0,2)$

What is the probability that the coin is at $(2,1)$?

$P(X=2, Y=1) = 0$

Finding a coin in the pond

Suppose you lost a coin in a 2 by 4 rectangular pond.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin → $X \sim U(0, 4)$

Let Y be the y-coordinate of the coin → $Y \sim U(0, 2)$

What is the probability that the coin is in the region where $0 \leq x \leq 2$ and $0.5 \leq y \leq 1$?

$$P(0 \leq X \leq 2,\ 0.5 \leq Y \leq 1) = \frac{(2-0)\times(1-0.5)}{4\times 2}.$$

$$= P(0 \leq X \leq 2)\, P(0.5 \leq Y \leq 1)$$

Finding a coin in the pond

Suppose you lost a coin in a 2 by 4 rectangular pond.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin $\longrightarrow$ $X \sim U(0, 4)$

Let Y be the y-coordinate of the coin $\longrightarrow$ $Y \sim U(0, 2)$

What is the probability that the coin is in the region where $0 \leq x \leq 2$ and $0.5 \leq y \leq 1$?

$$P(0 \leq X \leq 2) = \int_0^2 \frac{1}{4}\, dx = \frac{1}{2}$$

$$P(0.5 \leq Y \leq 1) = \int_{0.5}^1 \frac{1}{2}\, dx = \frac{1}{4}$$

Finding a coin in the pond

Suppose you lost a coin in a 2 by 4 rectangular pond.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin $\longrightarrow$ $X \sim U(0,4)$

Let Y be the y-coordinate of the coin $\longrightarrow$ $Y \sim U(0,2)$

What is the probability that the coin is in the region where $0 \le x \le 2$ and $0.5 \le y \le 1$?

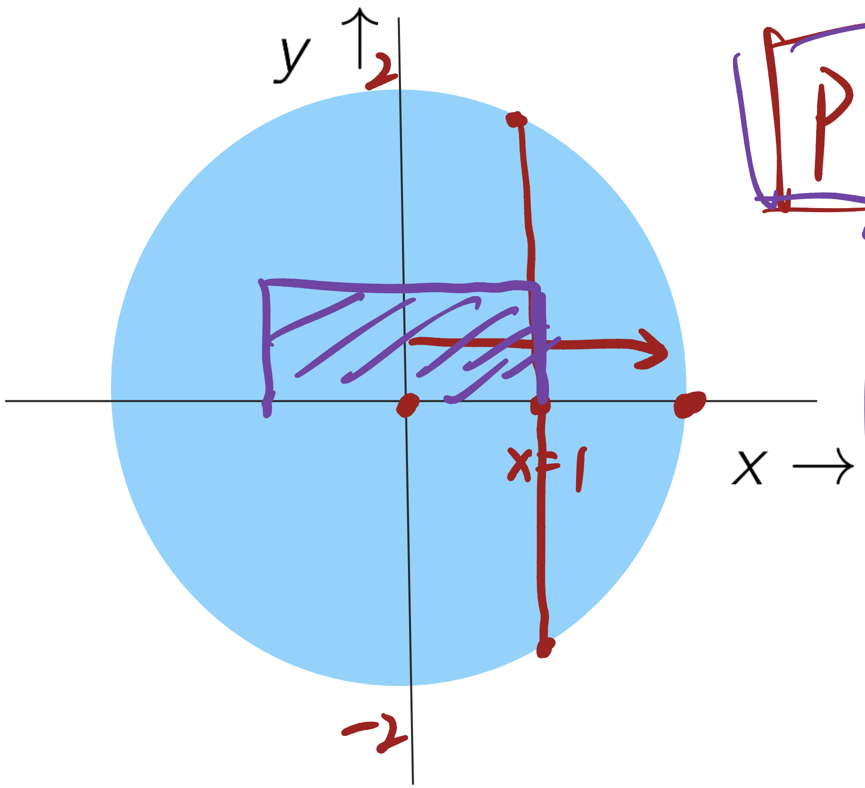
$$P(0 \le X \le 2) = \int_0^2 \frac{1}{4}\,dx = \frac{1}{2}$$

$$P(0.5 \le Y \le 1) = \int_{0.5}^1 \frac{1}{2}\,dx = \frac{1}{4}$$

$$\xrightarrow{\text{independence}} P(0 \le X \le 2, 0.5 \le Y \le 1) = \tfrac{1}{8}$$

Finding a coin in the pond

Suppose you lost a coin in a circular pond of radius 2.
You have no reason to suspect it in any particular location in the pond.



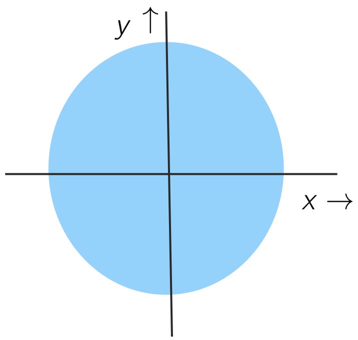
$$P(-1 \le X \le 1,\ 0 \le Y \le 1)$$

$$= P(-1 \le X \le 1) \times P(0 \le Y \le 1)$$

$$= \int_0^1 \int_{-1}^1 \underbrace{\frac{1}{4\pi}}_{\Delta A} \, dx\, dy.$$

$$\frac{2 \times 1}{4\pi}$$

Finding a coin in the pond



Suppose you lost a coin in a circular pond of radius 2.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin
Let Y be the y-coordinate of the coin

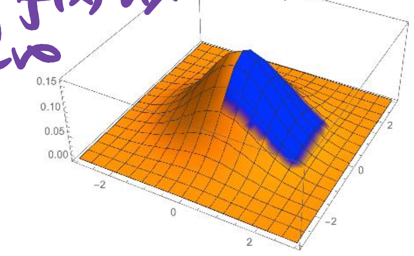
Neither are uniformly distributed!

The joint density function

$P(a \leq X \leq b) = \int_a^b f(x)\, dx$ → pdf

$f(x) \geq 0$

$\int_{-\infty}^{\infty} f(x)\, dx = 1$



Definition:

Let X and Y be two continuous RVs.

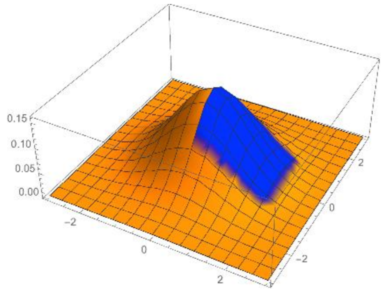
The joint density function f of X and Y is the function $f : \mathbb{R}^2 \to \mathbb{R}$ such that

$$P(a_1 \leq X \leq b_1, a_2 \leq Y \leq b_2) = \int_{a_2}^{b_2} \int_{a_1}^{b_1} f(x, y)\, dx dy.$$

1. $f(x, y) \geq 0$

2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1 = P(-\infty < X < \infty, -\infty < Y < \infty)$

The joint density function



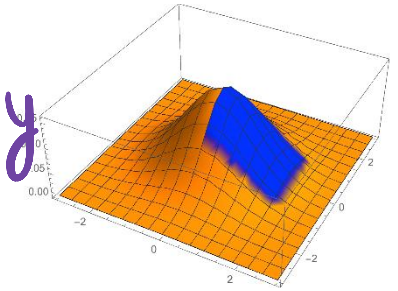
Definition:
Let X and Y be two continuous RVs.
The joint density function f of X and Y is the function $f : \mathbb{R}^2 \to \mathbb{R}$ such that

$$P(a_1 \leq X \leq b_1, a_2 \leq Y \leq b_2) = \int_{a_2}^{b_2} \int_{a_1}^{b_1} f(x, y)\, dx dy.$$

The density function has to satisfy

1. $f(x, y) \geq 0$;
2. $\displaystyle\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1.$

The joint density function



$$\int_{-1.5}^{1}\int_{-1.5}^{1} f(x,y)\,dx\,dy$$

PDF of X.

Volume.

Joint pdf of (X, Y).

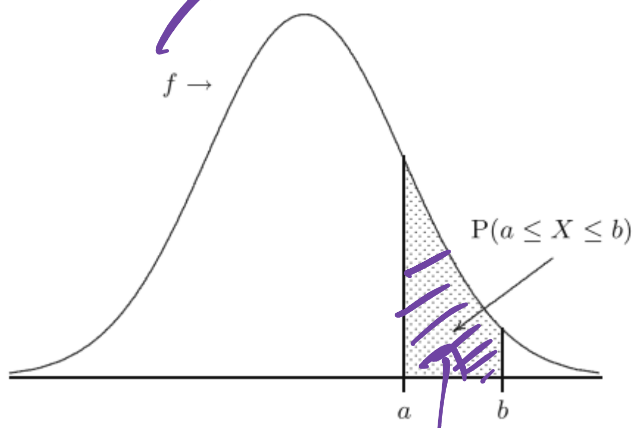


Fig. 5.1. Area under a probability density function f on the interval $[a, b]$.

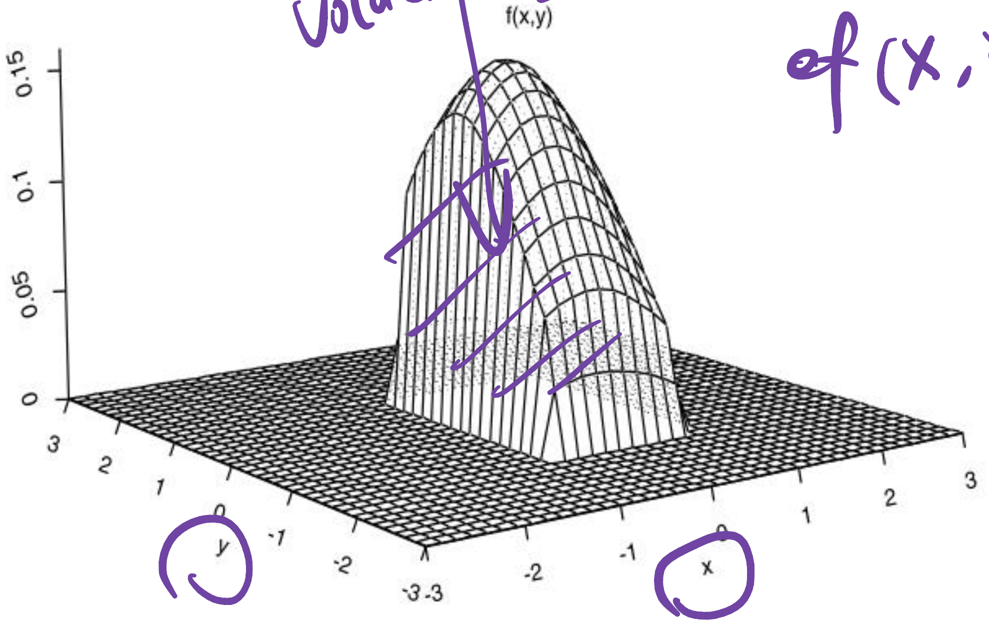


Fig. 9.1. Volume under a joint probability density function f on the rectangle $[-0.5, 1] \times [-1.5, 1]$.

$$P(a \le X \le b) = \boxed{\int_a^b f(x)\,dx}$$

area.

$$P(-0.5 \le X \le 1, -1.5 \le Y \le 1).$$

= Volume under $f(x,y)$.

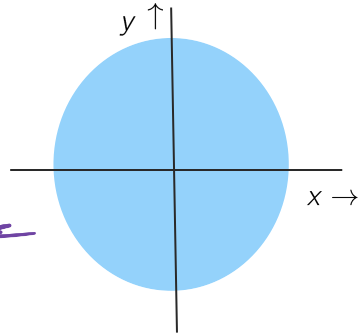
Finding a coin in the pond

Discrete. $p(x,y) \Rightarrow p_X(x) = \sum_y p(x,y)$

$p_Y(y) = \sum_x p(x,y)$

$Z \sim U(0,1)$

$f_Z(x) = 1, \; 0 \le x \le 1$



Suppose you lost a coin in a circular pond of radius 2.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin
Let Y be the y-coordinate of the coin

$$\boxed{f(x,y)} = \begin{cases} \frac{1}{4\pi} & \text{if } \sqrt{x^2+y^2} \le 2 \\ 0 & \text{otherwise} \end{cases}$$

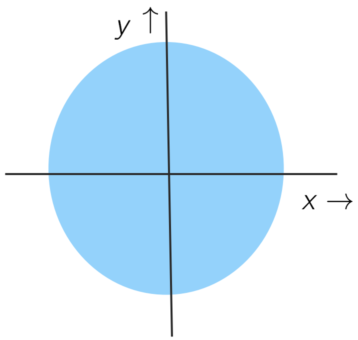
$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1$$

$$= \iint_{\sqrt{x^2+y^2}\le 2} f(x,y)\,dx\,dy \Rightarrow f(x,y) = \frac{1}{4\pi}$$

$$P(a \le X \le b, c \le Y \le d) = \int_c^d \int_a^b f(x,y)\,dx\,dy$$

$f(x,y) \Rightarrow f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy$ $\qquad f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx$

Finding a coin in the pond



Suppose you lost a coin in a circular pond of radius 2.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin $\longrightarrow$???
Let Y be the y-coordinate of the coin $\longrightarrow$???

$$f(x, y) = \begin{cases} \frac{1}{4\pi} & \text{if } \sqrt{x^2 + y^2} \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$P(a \leq X \leq b, c \leq Y \leq d) = \int_c^d \int_a^b f(x, y)\, dx\, dy$$

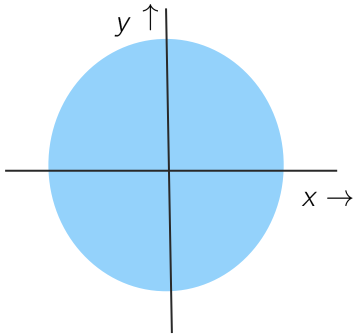
From joint to marginal density function

Let f be the joint density function of X and Y.
Then the marginal densities of X and Y can be found as

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy \qquad \text{and} \qquad f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx.$$

A A

Finding a coin in the pond



Suppose you lost a coin in a circular pond of radius 2.
You have no reason to suspect it in any particular location in the pond.

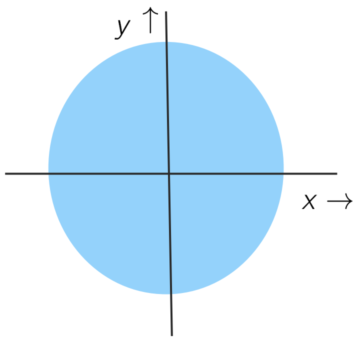
Let X be the x-coordinate of the coin → ???
Let Y be the y-coordinate of the coin → ???

$$f(x,y) = \begin{cases} \boxed{\frac{1}{4\pi}} & \text{if } \sqrt{x^2+y^2} \le 2 \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy = \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \frac{1}{4\pi}\,dy = \frac{\sqrt{4-x^2}}{2\pi}, \quad -2 \le x \le 2$$

$$f_Y(y) = \frac{\sqrt{4-y^2}}{2\pi}, \quad -2 \le y \le 2.$$

Finding a coin in the pond



Suppose you lost a coin in a circular pond of radius 2.
You have no reason to suspect it in any particular location in the pond.

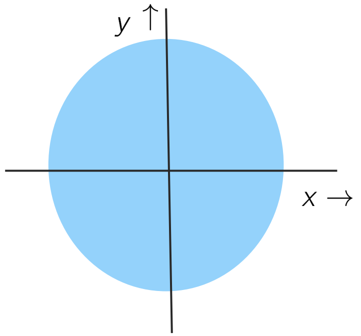
Let X be the x-coordinate of the coin ???

Let Y be the y-coordinate of the coin ???

$$f(x, y) = \begin{cases} \frac{1}{4\pi} & \text{if } \sqrt{x^2 + y^2} \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy = \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \frac{1}{4\pi}\, dy = \frac{\sqrt{4 - x^2}}{2\pi} \qquad \text{for } -2 \leq x \leq 2$$

Finding a coin in the pond



Suppose you lost a coin in a circular pond of radius 2.
You have no reason to suspect it in any particular location in the pond.

Let X be the x-coordinate of the coin $\longrightarrow$???

Let Y be the y-coordinate of the coin $\longrightarrow$???

$$f(x, y) = \begin{cases} \frac{1}{4\pi} & \text{if } \sqrt{x^2 + y^2} \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy = \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \frac{1}{4\pi}\, dy = \frac{\sqrt{4 - x^2}}{2\pi} \qquad \text{for } -2 \leq x \leq 2$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx = \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} \frac{1}{4\pi}\, dx = \frac{\sqrt{4 - y^2}}{2\pi} \qquad \text{for } -2 \leq y \leq 2$$

Connection between distribution and density function

$f(x) = \frac{dF(x)}{dx}$

$F(x,y) \Rightarrow f(x,y)$

Let f be the joint density function of continuous variables X and Y. Then the joint distribution function of X and Y can be found as

$$F(a,b) = \int_{-\infty}^{b} \int_{-\infty}^{a} f(x,y)\, dx dy.$$

$$F(a,b) = P(X \le a, Y \le b)$$

Connection between distribution and density function

Let f be the joint density function of continuous variables X and Y.
Then the joint distribution function of X and Y can be found as

$$F(a, b) = \int_{-\infty}^{b} \int_{-\infty}^{a} f(x, y) \, dx dy.$$

Let F be the joint distribution function of X and Y.
Then the joint density function of X and Y can be found as

$$f(x, y) = \frac{\partial^2}{\partial x \partial y} F(x, y).$$

$\partial_1 \partial_2$

Joint and marginal density functions

Let f be the joint density function of two continuous random variables X and Y.

Consider the following two statements:

1. $f_X(x) = \int_{-\infty}^{+\infty} f(x, y)\, dx \rightarrow dy$
2. $f_X(x) = \frac{\partial}{\partial x} F(x, y) \rightarrow$ a function of x and y

a function of x

Which of these statements is true?

- ✓ **A.** None
- **B.** Only 1
- **C.** Only 2
- **D.** Both

$f_X(x) = \frac{dF(x)}{dx}$

$f_X(x) = \int_y f(x,y)\, dy$

$= \int_y \frac{d^2 F(x,y)}{dx\, dy}\, dy \neq \frac{dF(x,y)}{dx}$

Exercise containing it all . . . .

Suppose the joint density of two RV's X and Y is given by

$$f(x,y) = 2e^{-x-2y}, \text{ for } x \geq 0,\ y \geq 0, \text{ and } \quad f(x,y) = 0 \quad \text{elsewhere.}$$

a. Find the joint distribution function $F(x,y)$.

$$F(x,y) = \int_{-\infty}^{y}\int_{-\infty}^{x} f(a,b)\, da\, db$$

$$= (e^{-x} - 1)(e^{-2y} - 1)$$

$x \geqslant 0,\ y \geqslant 0$

Exercise containing it all . . . .

Suppose the joint density of two RV's X and Y is given by

$$f(x,y) = 2e^{-x-2y}, \text{ for } x \geq 0,\ y \geq 0, \text{ and } \quad f(x,y) = 0 \quad \text{elsewhere.}$$

b. Find the marginal densities and distribution functions of X and Y.

$$f_X(x) = \int_y f(x,y)\,dy = e^{-x}$$

$$f_Y(y) = \int_x f(x,y)\,dx$$

$$F_X(x) = \int_{-\infty}^{x} f_X(a)\,da = \ldots$$

LOTUS for Multiple Random Variables

$Y = g(X)$ $\quad E(Y) = \int g(x) f(x)\, dx \Leftarrow$ LOTUS

No surprises in the definition of expectation or with LOTUS when more than one random variable is involved

$$E[g(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(u,v)\, f_{X,Y}(u,v)\, du\, dv$$

In particular

$$\boxed{E[aX + bY + c] = aE[X] + bE[Y] + c}$$

$g(X,Y) = aX + bY + c$ $\qquad$ linearity of expectation.

$E(aX) + E(bY) + c$

$= aE(X) + bE(Y) + c.$

Conditional PDF

The conditional pdf of Y given X is

$$\Rightarrow \int_{u} f_{Y|X}(v|u_0)\, dv = 1$$

$$f_{Y|X}(v|u_0) = \begin{cases} \dfrac{f_{X,Y}(u_0, v)}{f_X(u_0)}, & -\infty < v < \infty \\ \text{undefined}, & f_X(u_0) = 0 \end{cases}$$

Discrete case. conditional pmf.

$$P_{Y|X}(v|u_0) = \frac{P_{X,Y}(u_0, v)}{P_X(u_0)} \Rightarrow \sum_{v} P_{Y|X}(v|u_0) = 1$$

Connections with Conditional PDF

Conditional and marginal pdfs can be combined to yield a version of the law of total probability

$$f_{X,Y}(u,v) = f_{Y|X}(v \mid u)\, f_X(u)$$

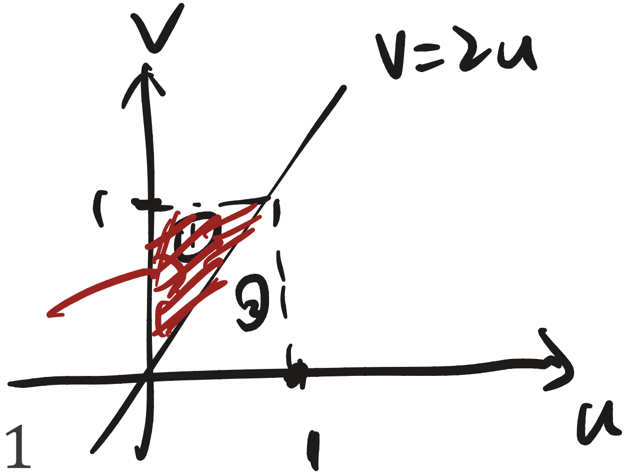
$$f_Y(v) = \int_{-\infty}^{\infty} f_{X,Y}(u,v)\, du$$

$$= \int_{-\infty}^{\infty} f_{Y|X}(v \mid u)\, f_X(u)\, du$$

The conditional expectation / mean of Y given $X = u$:

$$\underbrace{E[Y \mid X = u]}_{\boxed{\text{(a function of } u\text{)}}} = \int_{-\infty}^{\infty} v\, f_{Y|X}(v \mid u)\, dv$$

Example



The random variables (X, Y) are jointly distribution with

$$f_{X,Y}(u,v) = \begin{cases} u+v, & 0 \le u \le 1, 0 \le v \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Determine a) f_X, b) $f_{Y|X}$, and c) $P\{Y \ge 2X\}$ $= \iint_{\boxed{\{v \ge 2u\}}} f(u,v)\, du\, dv.$

a) The marginal distribution is given by

$$= \int_{\boxed{0}}^{\boxed{1}} \int_{\boxed{0}}^{\boxed{v/2}} f(u,v)\, du\, dv$$

$$\cancel{= \int_0^1 \int_{2u}^1 f(u,v)\, dv\, du}$$

Example

The random variables (X, Y) are jointly distribution with

$$f_{X,Y}(u,v) = \begin{cases} u+v, & 0 \le u \le 1, 0 \le v \le 1 \\ 0, & \text{otherwise} \end{cases}$$

Determine a) f_X, b) $f_{Y|X}$, and c) $P\{Y \ge 2X\}$

b) The conditional distribution is given by

Practice now

The joint probability density function f of the pair (X, Y) is given by

$$f(x, y) = K(3x^2 + 8xy) \text{ for } 0 \leq x \leq 1 \text{ and } 0 \leq y \leq 2,$$

and $f(x, y) = 0$ for all other values of x and y. Here K is some positive constant.

a. Find K.

b. Determine the probability $\boxed{P(2X \leq Y).}$

See you next week